

Q1: CSP – Graph Coloring with 4 Colors + Heuristics

Graph:

Nodes: {P, Q, R, S, T, U, V}

Edges:

- P–Q, P–R, P–S
- Q–R, Q–T
- R–S, R–U
- S–U, S–V
- T–U
- U–V

Tasks:

- Draw the graph clearly.
- Use colors {Red, Green, Blue, Yellow}.
- Solve using:
 - MRV
 - Degree Heuristic
 - Least Constraining Value (LCV)
- Show step-by-step assignment + domain pruning.

Q2: CSP – Scheduling with Global Constraints

You are scheduling 5 tasks {T1–T5} in time slots {1–5}.

Constraints:

- $T1 < T3 < T5$
- $T2 \neq T3$
- $T4 > T2$
- $T5 \neq 5$
- $|T1 - T2| \geq 2$

- All tasks must have different time slots

Tasks:

- Formulate as a CSP with global constraint (AllDiff).
- Apply Arc Consistency (AC-3) and show queue operations.
- Then use Backtracking with Forward Checking.
- Show intermediate domain reductions.

Q3: Cryptarithmic

Solve:

FORTY + TEN + TEN = SIXTY

Tasks:

- Identify all carry variables.
- Write full constraint system.
- Solve step-by-step.
- Clearly justify each deduction.